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MCY 611
Lab 5 AWS Auto Scaling

MCY 611 Cloud Computing Lab#5: AWS Auto Scaling

Part 1: Scale Up

1. Go to the directory where you keep the apache-install script file that you created in Lab 4.

2. Create a launch template (previously known as a launch configuration) using the following command:

```
aws ec2 create-launch-template --launch-template-name your_username-lt --version-description "my version" --launch-template-data '{ "ImageId": "ami-09e67e426f25ce0d7", "InstanceType":
```

```
"t3.micro", "KeyName": "your_username-key", "SecurityGroupIds":
```

```
["your_security_group_id"], "UserData": "$ (base64 -w 0 apache-install.sh)"}'
```

What is your command and its output?

```
~ $ aws ec2 create-launch-template --launch-template-name carrd12-lt --version-description "my version" --launch-template-data '{ "ImageId": "ami-09e67e426f25ce0d7", "InstanceType": "t3.micro", "KeyName": "carrd12-key", "SecurityGroupIds": ["sg-00c15e3626170b8b6"], "UserData": "$ (base64 -w 0 apache-install.sh)"}'
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0ed65c0001bb636c4",
    "LaunchTemplateName": "carrd12-lt",
    "CreateTime": "2025-02-14T16:30:18+00:00",
    "CreatedBy": "arn:aws:iam::881490118823:root",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      "Managed": false
    }
  }
}
~ $
```

3. Create a load balancer via the following command:

```
aws elb create-load-balancer --load-balancer-name your_username-elb --listeners
```

```
"Protocol=HTTP,LoadBalancerPort=80,InstanceProtocol=HTTP,InstancePort=80" --subnets
```

```
your_subnet_id --security-groups your_security_group_id
```

What is your command and its output?

```
~ $ aws elb create-load-balancer --load-balancer-name carrd12-elb --listeners "Protocol=HTTP,LoadBalancerPort=80,InstanceProtocol=HTTP,InstancePort=80" --subnets subnet-0aa6a451bcd4099f --security-groups sg-00c15e3626170b8b6
{
  "DNSName": "carrd12-elb-953447496.us-east-1.elb.amazonaws.com"
}
~ $
```

4. Create an auto scaling group via the following command:

```
aws autoscaling create-auto-scaling-group --auto-scaling-group-name your_username-asg
```

```
--launch-template "LaunchTemplateName=your_username-lt,Version=1" --min-size 1 --max-size 3
```

```
--vpc-zone-identifier "your_subnet_id"
```

What is your command?

```
~ $ aws autoscaling create-auto-scaling-group --auto-scaling-group-name carrd12-asg --launch-template "LaunchTemplateName=carrd12-lt,Version=1" --min-size 1 --max-size 3 --vpc-zone-identifier "subnet-0aa6a451bcd4099f"
~ $
```

5. Describe the auto scaling group via the following command:

```
aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name your_username-asg
```

What is the output? You should see one EC2 instance in your auto scaling group. If it doesn't appear, wait for 2 minutes and run the command again. **What is the instance ID?**

Instance ID i-04285157f456ee89c

```
~ $ aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name carrd12-asg
{
  "AutoScalingGroups": [
    {
      "AutoScalingGroupName": "carrd12-asg",
      "AutoScalingGroupARN": "arn:aws:autoscaling:us-east-1:881490118823:autoScalingGroup:ff806997-fbba-4a6e-b8ed-53433437c1ec:autoScalingGroupName/carrd12-asg",
      "LaunchTemplate": {
        "LaunchTemplateId": "lt-0ed65c001bb636c4",
        "LaunchTemplateName": "carrd12-lt",
        "Version": "1"
      },
      "MinSize": 1,
      "MaxSize": 3,
      "DesiredCapacity": 1,
      "DefaultCooldown": 300,
      "AvailabilityZones": [
        "us-east-1a"
      ],
      "LoadBalancerNames": [],
      "TargetGroupARNs": [],
      "HealthCheckType": "EC2",
      "HealthCheckGracePeriod": 0,
      "Instances": [],
      "CreatedTime": "2025-02-14T16:42:27.441000+00:00",
      "SuspendedProcesses": [],
      "VPCZoneIdentifier": "subnet-0aa6a451bcd4099f",
      "EnabledMetrics": [],
      "Tags": []
    }
  ]
}
```

6. Create a scale-up policy via the following command:

aws autoscaling put-scaling-policy --auto-scaling-group-name *your_username*-asg --policy-name *your_username*-scaleup --scaling-adjustment 1 --adjustment-type ChangeInCapacity --cooldown 120

What is your command and its output? What is the value of PolicyARN? (The value is a long string without quote characters). You need this value for next step.

PolicyARN

arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:06832dba-c8d7-497a-9e0a-90450288dfbb:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaleup

```
~ $ aws autoscaling put-scaling-policy --auto-scaling-group-name carrd12-asg --policy-name carrd12-scaleup --scaling-adjustment 1 --adjustment-type ChangeInCapacity --cooldown 120
{
  "PolicyARN": "arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:06832dba-c8d7-497a-9e0a-90450288dfbb:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaleup",
  "Alarms": []
}
~ $
```

7. Create an alarm to trigger the scale-up policy via the following command:

aws cloudwatch put-metric-alarm --alarm-name *your_username*-highcpualarm --metric-name CPUUtilization --namespace AWS/EC2 --statistic Average --period 120 --threshold 50 --comparison-operator GreaterThanThreshold --dimensions "Name=AutoScalingGroupName,Value=*your_username*-asg" --evaluation-periods 1 --alarm-actions *value_of_PolicyARN*

What is your command?

8. Describe the alarm via the following command:

aws cloudwatch describe-alarms --alarm-names *your_username*-highcpualarm

What is your command and its output?

```
~ $ aws cloudwatch put-metric-alarm --alarm-name carrd12-highcpualarm --metric-name CPUUtilization --namespace AWS/EC2 --statistic Average --period 120 --threshold 50
--comparison-operator GreaterThanThreshold --dimensions "Name=AutoScalingGroupName,Value= carrd12-asg" --evaluation-periods 1 --alarm-actions arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:06832dba-c8d7-497a-9e0a-90450288dfbb:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaleup
~ $ aws cloudwatch describe-alarms --alarm-names carrd12-highcpualarm
{
  "MetricAlarms": [
    {
      "AlarmName": "carrd12-highcpualarm",
      "AlarmArn": "arn:aws:cloudwatch:us-east-1:881490118823:alarm:carrd12-highcpualarm",
      "AlarmConfigurationUpdatedTimestamp": "2025-02-14T16:54:27.951000+00:00",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:06832dba-c8d7-497a-9e0a-90450288dfbb:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaleup"
      ],
      "InsufficientDataActions": [],
      "StateValue": "INSUFFICIENT_DATA",
      "StateReason": "Unchecked: Initial alarm creation",
      "StateUpdatedTimestamp": "2025-02-14T16:54:27.951000+00:00",
      "MetricName": "CPUUtilization",
      "Namespace": "AWS/EC2",
      "Statistic": "Average",
      "Dimensions": [
        {
          "Name": "AutoScalingGroupName",
          "Value": "carrd12-asg"
        }
      ]
    }
  ]
}
```

9. Log in to the EC2 instance created in Step 5 via SSH. **What are your commands and their outputs?** Once logged in, this CloudShell terminal will become an EC2 instance terminal. Switch to the root user. **What is your command and its output?**

```
~ $ ssh -i devenv-key.pem ubuntu@100.29.9.176
The authenticity of host '100.29.9.176 (100.29.9.176)' can't be established.
ED25519 key fingerprint is SHA256:SftJTN8Rs4XxY+1JyVY+19/RqcXwnTjo1Gc0UueVgkw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.29.9.176' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-1045-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Feb 14 16:58:13 UTC 2025

System load:  0.0           Processes:      114
Usage of /:   25.8% of 7.69GB Users logged in: 0
Memory usage: 26%          IPv4 address for ens5: 172.31.32.234
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.

ubuntu@ip-172-31-32-234:~$ sudo -s
root@ip-172-31-32-234:/home/ubuntu#
```

Install a stress Linux tool via the following command:

apt install stress -y

What is the output? The stress command is used to test system performance by generating load on various system resources such as CPU, memory, and disk.

```

root@ip-172-31-32-234:/home/ubuntu# apt install stress -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupdplugin1 libxmlb1
Use 'apt autoremove' to remove them.
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 55.3 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal/universe amd64 stress amd64 1.0.4-6 [18.4 kB]
Fetched 18.4 kB in 0s (230 kB/s)
Selecting previously unselected package stress.
(Reading database ... 90405 files and directories currently installed.)
Preparing to unpack .../stress_1.0.4-6_amd64.deb ...
Unpacking stress (1.0.4-6) ...
Setting up stress (1.0.4-6) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for man-db (2.9.1-1) ...
root@ip-172-31-32-234:/home/ubuntu#

```

Run the following command to generate workload so that you can see the scale-up process:

`stress --cpu 8`

This option spawns 8 worker threads, each fully utilizing one CPU core of your EC2 instance. **What is the output?**

```

root@ip-172-31-32-234:/home/ubuntu# stress --cpu 8
stress: info: [38706] dispatching hogs: 8 cpu, 0 io, 0 vm, 0 hdd

```

10. **Start a new CloudShell terminal.** Run the following command every 2 minutes until you see two EC2 instances in your Auto Scaling group. If you wait longer, you may see three EC2 instances, but it's not necessary to wait for the third one. You can proceed to the next step as soon as two EC2 instances are present in your auto scaling group.

`aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name your_username-asg`

What is the output?

```

~ $ aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name carrd12-asg
{
  "AutoScalingGroups": [
    {
      "AutoScalingGroupName": "carrd12-asg",
      "AutoScalingGroupARN": "arn:aws:autoscaling:us-east-1:881490118823:autoScalingGroup:ff806997-fbba-4a6e-b8ed-53433437c1ec:autoScalingGroupName/carrd12-asg",
      "LaunchTemplate": {
        "LaunchTemplateId": "lt-0ed65c001bb636c4",
        "LaunchTemplateName": "carrd12-1t",
        "Version": "1"
      },
      "MinSize": 1,
      "MaxSize": 3,
      "DesiredCapacity": 2,
      "DefaultCooldown": 300,
      "AvailabilityZones": [
        "us-east-1a"
      ],
      "LoadBalancerNames": [],
      "TargetGroupARNs": [],
      "HealthCheckType": "EC2",
      "HealthCheckGracePeriod": 0,
      "Instances": [
        {
          "InstanceId": "i-024bae8037e33501d",
          "InstanceType": "t3.micro",
          "AvailabilityZone": "us-east-1a",
          "LifecycleState": "InService"
        }
      ]
    }
  ]
}

```

Part 2: Scale Down

11. Create a scale-down policy via the following command:

aws autoscaling put-scaling-policy --auto-scaling-group-name *your_username*-asg --policy-name *your_username*-scaledown --scaling-adjustment -1 --adjustment-type ChangeInCapacity --cooldown 120
What is your command and its output? What is the value of PolicyARN? (The value is a long string without quote characters). You need this value for next step.

```
~ $ aws autoscaling put-scaling-policy --auto-scaling-group-name carrd12-asg --policy-name carrd12-scaledown --scaling-adjustment -1 --adjustment-type ChangeInCapacity --cooldown 120
{
  "PolicyARN": "arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:ae88279b-1867-4550-9ca5-5bb9c221d2c6:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaledown",
  "Alarms": []
}
```

12. Create an alarm to trigger the scale-down policy via the following command:

aws cloudwatch put-metric-alarm --alarm-name *your_username*-lowcpualarm --metric-name CPUUtilization --namespace AWS/EC2 --statistic Average --period 120 --threshold 30 --comparison-operator LessThanThreshold --dimensions "Name=AutoScalingGroupName,Value=*your_username*-asg" --evaluation-periods 1 --alarm-actions *value_of_PolicyARN*

What is your command?

```
~ $ aws cloudwatch put-metric-alarm --alarm-name carrd12-lowcpualarm --metric-name CPUUtilization --namespace AWS/EC2 --statistic Average --period 120 --threshold 30 --comparison-operator LessThanThreshold --dimensions "Name=AutoScalingGroupName,Value=carrd12-asg" --evaluation-periods 1 --alarm-actions arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:ae88279b-1867-4550-9ca5-5bb9c221d2c6:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaledown
~ $ aws cloudwatch describe-alarms --alarm-names carrd12-lowcpualarm
{
  "MetricAlarms": [
    {
      "AlarmName": "carrd12-lowcpualarm",
      "AlarmArn": "arn:aws:cloudwatch:us-east-1:881490118823:alarm:carrd12-lowcpualarm",
      "AlarmConfigurationUpdatedTimestamp": "2025-02-14T17:16:41.751000+00:00",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:autoscaling:us-east-1:881490118823:scalingPolicy:ae88279b-1867-4550-9ca5-5bb9c221d2c6:autoScalingGroupName/carrd12-asg:policyName/carrd12-scaledown"
      ],
      "InsufficientDataActions": [],
      "StateValue": "OK",
      "StateReason": "Threshold Crossed: 1 datapoint [45.65613302795853 (14/02/25 17:14:00)] was not less than the threshold (30.0).",
      "StateReasonData": "[{\"version\":\"1.0\",\"queryDate\":\"2025-02-14T17:18:07.265+0000\",\"startDate\":\"2025-02-14T17:14:00.000+0000\",\"statistic\":\"Average\",\"period\":\"120\",\"recentDatapoints\":[45.65613302795853],\"threshold\":\"30.0\",\"evaluatedDatapoints\":[{\"timestamp\":\"2025-02-14T17:14:00.000+0000\",\"sampleCount\":\"8.0\",\"value\":\"45.65613302795853\"}]]",
      "StateUpdatedTimestamp": "2025-02-14T17:18:07.269000+00:00",
      "MetricName": "CPUUtilization",
      "Namespace": "AWS/EC2",
      "Statistic": "Average",
      "Dimensions": [
        {
          "Name": "AutoScalingGroupName"
        }
      ]
    }
  ]
}
```

13. **Switch to the EC2 instance terminal.** Please type Ctl-C to stop the stress command (workload) so that you can see the scale-down process.

14. **Switch back to the CloudShell terminal.** Repeat the following command every 2 minutes until you see only one EC2 instance in your auto scaling group:

aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name *your_username*-asg

What is the output?

```

~ $ aws autoscaling describe-auto-scaling-groups --auto-scaling-group-name carrd12-asg
{
  "AutoScalingGroups": [
    {
      "AutoScalingGroupName": "carrd12-asg",
      "AutoScalingGroupARN": "arn:aws:autoscaling:us-east-1:881490118823:autoScalingGroup:ff806997-fbba-4a6e-b8ed-53433437c1ec:autoScalingGroupName/carrd12-asg",
      "LaunchTemplate": {
        "LaunchTemplateId": "lt-0ed65c0001bb636c4",
        "LaunchTemplateName": "carrd12-1t",
        "Version": "1"
      },
      "MinSize": 1,
      "MaxSize": 3,
      "DesiredCapacity": 1,
      "DefaultCooldown": 300,
      "AvailabilityZones": [
        "us-east-1a"
      ],
      "LoadBalancerNames": [],
      "TargetGroupARNs": [],
      "HealthCheckType": "EC2",
      "HealthCheckGracePeriod": 0,
      "Instances": [
        {
          "InstanceId": "i-024bae8037e33501d",
          "InstanceType": "t3.micro",
          "AvailabilityZone": "us-east-1a",
          "LifecycleState": "Terminating",
          "HealthStatus": "Healthy"
        }
      ]
    }
  ]
}

```

Part 3: Cleanup

15.Delete the auto scaling group via the following command. If the group has instances or scaling activities in progress, you must specify the option to force the deletion in order for it to succeed. If the group has policies, deleting the group deletes the policies, the underlying alarm actions, and any alarm that no longer has an associated action.

aws autoscaling delete-auto-scaling-group --auto-scaling-group-name *your_username*-asg --force-delete

16.Delete both alarms via the following commands:

aws cloudwatch delete-alarms --alarm-name *your_username*-lowcpualarm

aws cloudwatch delete-alarms --alarm-name *your_username*-highcpualarm

```

~ $ aws autoscaling delete-auto-scaling-group --auto-scaling-group-name carrd12-asg --force-delete
~ $ aws cloudwatch delete-alarms --alarm-name carrd12-lowcpualarm
~ $ aws cloudwatch delete-alarms --alarm-name carrd12-highcpualarm

```

17.Delete your config via the following command:

aws ec2 delete-launch-template --launch-template-name *your_username*-lt

What is the output?

```

~ $ aws ec2 delete-launch-template --launch-template-name carrd12-1t
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0ed65c0001bb636c4",
    "LaunchTemplateName": "carrd12-1t",
    "CreateTime": "2025-02-14T16:30:18+00:00",
    ...skipping...
  },
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0ed65c0001bb636c4",
    "DefaultVersionNumber": 1,
    ...skipping...
  },
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0ed65c0001bb636c4",
    "LaunchTemplateName": "carrd12-1t",
    "CreateTime": "2025-02-14T16:30:18+00:00",
    "CreatedBy": "arn:aws:iam:881490118823:root",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      ...skipping...
    }
  },
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0ed65c0001bb636c4",
    "LaunchTemplateName": "carrd12-1t",
    "CreateTime": "2025-02-14T16:30:18+00:00",
    "CreatedBy": "arn:aws:iam:881490118823:root",
    ...skipping...
  }
}

```

18.Delete the ELB. **What command did you run?**

```
~ $ aws elb delete-load-balancer --load-balancer-name carrd12-elb
~ $
```

19. Delete your key pair with the following command:

aws ec2 delete-key-pair --key-name *your_username*-key

```
~ $ aws ec2 delete-key-pair --key-name carrd12-key
{
  "Return": true,
  "KeyPairId": "key-02c03f4a24005e10f"
}
~ $
```

Submission instruction

After completing this work, submit this word document with your answers to the Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

Studying Amazon Auto Scaling and learning how to dynamically manage EC2 instances in response to demand were the objectives of this lab. To achieve high availability, performance, and cost effectiveness, I set up Auto Scaling using Amazon EC2 and Elastic Load Balancing (ELB) using the AWS CLI. I started by defining the EC2 instance setup using a Launch Template and an Auto Scaling group in order to accomplish these objectives. After that, I configured scaling policies to automatically change the number of instances in response to incoming traffic or CPU usage. In order to uniformly distribute traffic among healthy instances, I then combined Elastic Load Balancing (ELB) with the Auto Scaling group. In order to keep an eye on instance health and initiate scaling events as needed, I also set up Amazon CloudWatch alarms. In order to see how the Auto Scaling team automatically changed the quantity of active EC2 instances in response to demand, I lastly tested the configuration by creating traffic.